BIRDB: Building an Affordable and Metadata-hiding Encrypted Database via Compact Encrypted Search

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***Abstract*—The escalating concerns surrounding the security of data outsourced to untrusted cloud servers have driven significant advancements in *Encrypted Databases* (EDB). EDBs provide secure storage, search, and processing of encrypted data while ensuring the confidentiality of search metadata. However, existing approaches face challenges in achieving both efficient encrypted search and metadata confidentiality. One line of work constructs cryptographic indexes on property-preserving ciphertexts for efficient searches but suffers from a substantial increase in index size and inadvertent metadata leakage. The other line of work resorts to scanning the entire dataset to hide metadata, resulting in significant overhead for encrypted search.**

We present BIRDB, the first high-performance and metadata-hiding EDB that offers an affordable encrypted search solution. At the core of BIRDB lies a novel *boolean information retrieval* technique, enabling logarithmic search complexity on ciphertexts within a searchable index structure. The boolean nature of the index provides two crucial advantages: (1) lossless index compression through linear approximation and (2) seamless integration with cutting-edge private information retrieval (PIR) protocols. This integration allows computationally expensive query-independent computations to be shifted to an offline phase, leading to sublinear computation during the query-specific online phase, all while ensuring metadata protection. Extensive evaluations demonstrate the superiority of BIRDB, achieving up to 15.2X higher query throughput, occupying 1018X less space, and providing robust metadata protection compared to existing notable EDB solutions.

I. INTRODUCTION

The popularity of cloud computing has prompted numerous organizations to adopt cloud storage services like DropBox [28] and Onedrive [80] for their scalability and rich query capabilities [9], [43], [81]. However, this increased reliance has also raised significant security concerns. For example, a recent data breach resulted in the exposure of personal information belonging to 237,000 US government employees [1]. Samsung has leaked its confidential source code, where employees had uploaded it to ChatGPT for solutions [7]. TikTok has also faced accusations of compromising user privacy, particularly regarding the data of over 150 million US citizens uploaded to the cloud [8].

Recently, *encrypted databases* (EDBs) have emerged as a reliable solution for confidential computing. By storing data solely in encrypted form, without decryption keys, EDBs ensure data confidentiality against privileged database administrators and external attackers (adversaries). To enable secure query processing, EDBs must maintain searchability for large and constantly growing encrypted data sets. Additionally, they employ oblivious algorithms to protect search metadata, such as data access footprint, preventing adversaries from extracting sensitive information about the encrypted data [21], [30], [85].

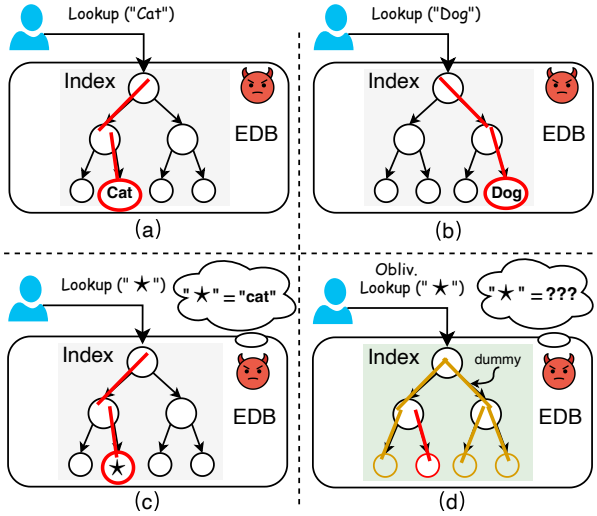


Fig. 1: Compared to an EDB that uses only end-to-end encryption shown in **(a)**∼**(c)**, a metadata-hiding EDB in **(d)** disguises intended index operations with dummy operations, effectively preventing adversaries from determining which index node is being accessed, ensuring enhanced confidentiality protection.

Consider an example of how an adversary can exploit search access patterns to exfiltrate data and how an EDB would benefit from metadata protection [24]. Figure 1 depicts an EDB server that stores a search index, typically a binary search tree (BST) or its variants, that maps an encrypted keyword to its file. The adversary first observes a series of plaintext queries and their respective index operations shown in (a)∼(b). In this example, it observes two point lookup queries retrieving files containing the keywords 'cat' and 'dog' and their respective search paths. By repeating this process, the adversary can learn the keyword linked to every index node; subsequently, when a user submits a confidential query with an encrypted search keyword ('⋆' in (c)), the adversary can deduce the keyword without requiring decryption keys based on the search path and the specific index node accessed, causing unexpected data leakage. An EDB with metadata protection employs oblivious algorithms that incorporate dummy accesses ((**d**)), to conceal the actual search access patterns, thus mitigating the search pattern vulnerability.

A commonly employed approach to achieve both searchability and metadata protection is the use of Oblivious RAM (ORAM) [29], [62], [73]. ORAM-based methods enable cryptographic concealment of read/write operations on servers while keeping search accesses confidential. Several academic studies have proposed incorporating a BST within ORAM as a straightforward technique [27], [34], [71]. However, despite the polylogarithmic asymptotic complexity of ORAM in relation

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to the size of the index, the practical implementation of ORAM schemes remains prohibitively expensive. This is primarily due to the requirement of executing all oblivious computations within a single query-processing stage [24], [62].

As an alternative, the second line of work leverages specific cryptographic tools to maintain either searchability or metadata privacy. Specifically, CryptDB [64] and its variants [38], [76] use property-preserving encryption algorithms [15], [10]. These algorithms maintain determinism and ordering of encrypted data, enabling the construction of BSTs with logarithmic search complexity. However, these approaches tend to overlook metadata security. Additionally, they often suffer from unexpectedly poor query performance due to the substantial increase in the size of encrypted index nodes because this surged index size necessitates a higher reliance on I/O expensive external storage solutions such as disks [31], [42].

To address the challenge of increased index size in ciphertexts and improve query performance, HELibDB [22] employs Fermat's Little Theorem [79] to directly assess the equivalence of homomorphically encrypted data [32], [87], eliminating the need for index construction. However, this direct evaluation necessitates linearly scanning a large amount of encrypted data and consumes substantial bandwidth due to multi-round communication requirements. Another approach, OO-PIR [53], preprocesses the database into multiple chunks, where each online query retrieves an entire chunk as dummy operations to conceal the accessed data. Although costly oblivious operations are moved to an offline phase, it still requires scanning each chunk to test the membership of the queried data, resulting in a significant decrease in performance.

Overall, the development of an EDB that can achieve efficiency (logarithmic search complexity), compactness (constant space for indexing regardless of the total stored data), and metadata-hiding is still an open problem and highly desirable.

Drawing inspiration from existing Boolean Information Retrieval (BIR) protocols developed for document retrieval [12], [48], our key observation is that the *membership* information has significant potential to achieve all the aforementioned properties simultaneously. Specifically, we identify two distinct benefits of membership information. Firstly (searchability), it provides a new approach to enable search on ciphertexts by replacing traditional numerical comparisons in BSTs with membership tests. Secondly (compactness), the boolean characteristic of membership information allows for the integration of advanced compression techniques [45], [49]. This addresses the inherent inefficiency of indexing ciphertexts as the storage space required for indexing ciphertexts can be reduced. Moreover, this approach aligns well with recent advancements in two-phase PIR [39], [53], where the database is preprocessed using boolean operations such as XOR. Overall, this indicates compatibility and potential synergy of using one boolean abstraction to achieve all properties in a unified manner.

This paper introduces BIRDB[1], a high-performance encrypted database with metadata protection. At the core of BIRDB is a novel secure and compact index called BINDEX. Unlike previous approaches that employ ORAM-based or cryptographic methods resulting in uncompressible indexes

[1]BIRDB stands for **B**oolean **I**nformation **R**etrieval based E**DB**.

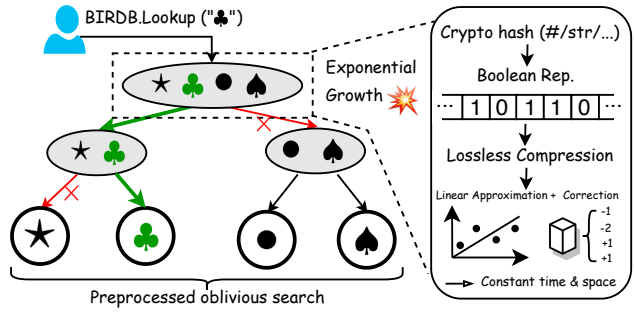


Fig. 2: BIRDB's fundamental concept revolves around leveraging Boolean information to construct a searchable and compact index that incorporates an oblivious PIR protocol.

due to random noise introduced during encryption, BINDEX utilizes efficient one-way functions (OWF), such as SHA-256 [37], to transform data into computationally-irreversible ciphertexts within a membership-preserving data structure. The membership information associated with these ciphertexts is then leveraged for searching, while the boolean nature of index nodes allows for compressibility.

Figure 2 illustrates the fundamental concept of BINDEX. In this process, each item undergoes transformation using OWFs to generate *cipher identifiers*. These identifiers are organized within a carefully designed data structure that preserves membership information. Notably, even with knowledge of these identifiers, a computational bounded adversary cannot reverse-engineer the original plaintext because of the inherent guarantees provided by OWFs [37].

To achieve searchability, all cipher identifiers are positioned at the bottom layer of the index, and BINDEX is constructed by recursively combining them into a data structure resembling a Merkle tree. Each inner node in the structure represents the union of its child identifiers. The searchability is facilitated by BINDEX propagating the membership information of identifiers from the bottom layer to the root. This enables efficient top-down binary search by testing the membership of the queried item in each node. This search mechanism resembles the behavior of a BST commonly used in plaintext databases but without the need for numerical comparisons on plaintext values.

A key performance challenge of BINDEX lies in the space efficiency. Specifically, if the identifiers in index nodes are stored uncompressed, the space consumption would theoretically be proportional to the size of the dataset. Moreover, the unique leaf-to-root membership propagation, which enables searching without plaintexts, amplifies the space occupancy even further. This is due to the fact that inner nodes in BINDEX need to maintain the membership information of all child nodes, resulting in an exponentially growing index size.

To address this challenge, BIRDB implements a compact index design that incorporates lossless compression techniques for the top-level index nodes containing a significant amount of membership information from child nodes. Taking inspiration from the recent advancements in the field of linear approximation model [44], [45], [49], BIRDB apply offsets to represent the boolean information of large BINDEX nodes and employ linear approximation with corrections to map the keys in the input set to their respective positions, resulting in

compressed nodes with constant space occupancy. By doing so, the performance penalty arising from the expanded index on ciphertext is significantly reduced, making BIRDB's queries can run at high performance without compromising accuracy or security.

The boolean nature of BINDEX allows for seamless integration with state-of-the-art Private Information Retrieval (PIR) protocols, ensuring efficient metadata security. In practice, BINDEX is constructed and deployed on both an online EDB server and an offline server (e.g., a replica of the EDB). The offline server performs database preprocessing to generate auxiliary hints, which are then leveraged during the online phase to provide efficient responses to user queries with sublinear computation and communication [23], [53]. This integration is seamless because the preprocessing step also relies on boolean operations, typically XOR. The integration between BINDEX and the PIR protocol is mutually beneficial, as the searchable BINDEX accelerates the PIR process, while the PIR protocol ensures metadata hiding for BINDEX.

We implemented BIRDB on CryptDB's codebase [64]. BIRDB's modular architecture allows for the transparent deployment of decoupled index and storage optimizations with reasonable engineering efforts. To evaluate its performance, we compared BIRDB against CryptDB [64], HElibDB [22], and OO-PIR [53], which are two efficient EDBs for oblivious information retrieval, within the same system. We conducted evaluations using a typical workload from previous studies [88]. Evaluation shows that:

• BIRDB provides strong security. We prove the cryptographic confidentiality and metadata privacy guarantees in §**??**.

• BIRDB demonstrates high efficiency. BIRDB exhibits up to 33.5% lower query latency and 5.1X higher throughput than HElibDB, as well as up to 3.8% lower latency and 2.3X higher throughput than OO-PIR. These performance gains can be attributed to the logarithmic complexity and compactness of BINDEX (§V).

• BIRDB offers compactness. In terms of storage space, a compressed BINDEX occupies approximately 1018X less space than an uncompressed version while maintaining the same query accuracy (§V).

Our main contribution is BINDEX, the first high-performance and compact index on ciphertexts. BIRDB makes the first effective attempt to use boolean information for constructing a searchable and compressable index, and seamlessly integrates with state-of-the-art PIR protocol that preprocesses database using boolean operations, in one unified system. BIRDB has potential to be deployed in many safety-critical applications, including privacy-preserving video streaming [63], [84], password manager [33], recommendation [40], [68], and anonymous messaging [41], [67].

In the rest of this paper: §II introduces background; §III gives an overview of BIRDB; §IV describes BIRDB's design; §**??** covers BIRDB's security analysis; §V shows our evaluation; §VI discusses related work and §VII concludes.

II. BACKGROUND

In this section, we aim to highlight the importance of meeting three key properties - *searchability*, *compactness*, and *metadata protection* - in encrypted databases (compared in Table I). Understanding the significance of these properties will provide motivation for our design of BIRDB.

System	Data Confidentiality	Metadata Security	Sublinear Complexity	Compact Index	General Purpose
MySQL [56]	✗	✗	✓	✓	✓
CryptDB [64]	✓	✗	✓	✗	✓
HElibDB [22]	✓	✓	✗	✓	✓
Dory [25]	✓	✓	✗	✗	✗
OO-PIR [53]	✓	✓	✗	✗	✓
Titanium [53]	✓	✓	✗	✗	✓
BIRDB	✓	✓	✓	✓	✓

TABLE I: Characterization of representative EDBs.

A. Toward efficient server-side search

In response to growing security concerns about data leakage, enterprises like Keybase [5], PreVeil [6], SpiderOak [2], Sync [3], and Tresorit [4] have emerged, offering encrypted storage services to ensure user data confidentiality. These companies have expressed a need for server-side search functionality on encrypted data, in a way similar to conventional plaintext databases like MySQL [56]. However, none of these companies currently support server-side search, as their development efforts have been hindered by performance or the disclosure of search access patterns.

One straightforward yet inefficient way to enable search on encrypted data is to download the entire database. However, this approach is impractical due to the significant bandwidth consumption and client-side storage resources required. Thus although it provides the highest level of security, no existing industrial solution utilizes this approach.

To enable efficient server-side search on encrypted data, extensive academic research has explored the use of specific cryptographic tools, such as property-preserving encryption, as seen in the case of CryptDB [64]. The core concept behind this approach is to preserve the determinism and maintain the order of encrypted data, thereby allowing for the direct construction of an index (typically a BST [20]) on top of the encrypted data with logarithmic search complexity.

The need for compacting index. Despite extensive efforts to achieve efficient search performance, evaluations have shown that the actual performance often falls short of theoretical (i.e., logarithmic) expectations. The primary reason for this discrepancy is that the performance bottleneck does not lie in the binary search itself. Instead, it is the I/O operations to external storage that significantly degrade the overall performance. This is particularly true when dealing with substantial data volumes that cannot be indexed and fit within the limited main memory, compounded by the fact that encryption algorithms typically expand ciphertext size linearly [11], [74].

Therefore, in the context of EDB, achieving high performance necessitates not only sublinear (logarithmic) search complexity but also effective compression techniques. By ensuring sublinear search complexity, the search operations can be executed efficiently, while compression helps mitigate the impact of I/O operations and reduces the storage requirements.

B. Toward efficient metadata protection

Private Information Retrieval (PIR). The initial concept of Private Information Retrieval (PIR) was introduced by Chor et al. [57], allowing clients to retrieve specific objects from an untrusted server-managed database without revealing the accessed object. PIR protocols aim to minimize communication costs, achieving sublinear complexity relative to the database size. Modern PIR deployment models include multi-server PIR [26], [47], [50], [52], where the database is replicated across non-colluding servers for enhanced fault tolerance. Alternatively, single-server PIR protocols [54], [55], [60] are computationally more demanding but avoid non-collusion assumptions through cryptographic hardness assumptions. These advancements provide a range of options for secure and private information retrieval, considering efficiency, security assumptions, and deployment requirements.

Oblivious RAM (ORAM) is a traditional PIR technique known for concealing access patterns during interactions with untrusted servers [61], [72], however, it is not suitable for our setting due to its impractical performance penalty. ORAM protocols achieve access pattern obfuscation through techniques like oblivious data shuffling and path encryption. Researchers have also proposed various ORAM variants like Titanium [21] and Dory [24] to improve performance and scalability while preserving data privacy [17], [36], [66], [70], [71]. However, ORAM-based techniques still suffer from the drawback of processing the entire database with linear computation for every query, resulting in significant performance penalties.

Two-phase PIR on the boolean information. Many research efforts have been put on improving the performance of ORAM by introducing preprocessing techniques [14], [23], [69], [86]. Preprocessing involves shifting the inevitable PIR computation to a query-independent offline phase and generating auxiliary data or hints along the way. This approach aims to make the computational cost of online query processing, with the assistance of these hints, sublinear to the database size.

Two recent PIR protocols, namely the Corrigan-Gibbs and Kogan scheme (CK) [23], and the Shi, Aqeel, Chandrasekaran, and Maggs scheme (SACM) [69], operate on the boolean information to enable a two-phase design. Notably, the CK protocol [23] presents a two-server PIR construction that stores $\mathcal{O}(\sqrt{n})$ bits at the client and achieves $\mathcal{O}(\sqrt{n})$ server computation for online queries. The fundamental idea behind CK is the reflexivity of XOR operations. The CK protocol divides the database into random chunks, and the user preserves a preprocessed hint $h = p \oplus s$, where p represents the desired value to be queried and s is obtained by querying a random chunk of the database. Through this approach, the user can ultimately obtain $p = h \oplus s$ using its hint and queried random result, with correctness guaranteed by the reflexivity property (i.e., $p \oplus s \oplus s = p$).

C. Opportunities

Based on the survey results discussed above, we have identified several opportunities to address the challenge of practical yet secure encrypted search:

• Traditional cryptographic approaches are not capable of simultaneously achieving both searchability and compactness.

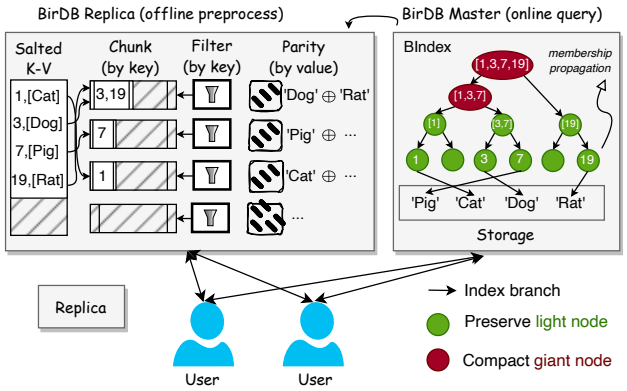


Fig. 3: BIRDB's architecture for searchable, compact, and metadata-hiding encrypted database built on the two-phase PIR.

Hence, there is a strong need for an alternative approach that offers a more flexible and practical index design, even if it entails sacrificing some degree of confidentiality. This trade-off is crucial to ensure the compactness of the encrypted search system, making it efficient for real-world applications.

• Boolean information has shown significant advancements in efficient metadata protection and holds great potential for implementation in an encrypted search index.

These opportunities serve as the basis for BIRDB's design.

III. Overview

A. System model

Same as [24], in BIRDB, the database items are divided into partitions, each managed by a different group of EDB servers. Multiple partitions can exist in a query system, and execution across partitions occurs in parallel.

Entities. For a single partition (depicted in Figure 3), we adopt the two-phase PIR model, specifically the CK protocol introduced by Corrigan-Gibbs and Kogan [23]. In our setup, a single partition consists of two servers deployed on the public cloud: an offline server (usually a replica) responsible for database preprocessing to generate runtime hints for PIR, and an online server (master) handling user queries. These servers are *semi-honest*, in the sense that they do not collude but have an interest in learning which objects the client is accessing from the database. Concretely, BIRDB's architecture for a single partition includes the following entities:

• **Users:** Multiple users interact with the BIRDB replica to obtain hints and send search queries (e.g., point lookup) to the BIRDB master for information retrieval. Each user needs to store a *salt*, a randomly generated 128-bit key, used for salting and recovering the original data during offline preprocessing to ensure confidentiality.

• **Master:** The BIRDB master processes user queries by searching BINDEX sublinearly and performing sublinear PIR computations. The master also ensures that the l replicas have a consistent view of the database state, and users know which servers to contact. In line with [24], we employ $l = 2$ in our implementation.

• **Replicas:** The BIRDB replica conducts database preprocessing to generate runtime *hints*, ensuring metadata-hiding during

information retrieval while keeping the computation required on the master sublinear.

Similar to [24], BIRDB does not provide availability if servers refuse to provide services. However, it is important to note that cloud providers are typically incentivized to ensure availability by either bringing the failed server back online or replacing it promptly [24], [43], [81].

Data model. To exemplify BIRDB's design, we employ the NoSQL key-value data model for its simplicity. However, it's important to note that BIRDB is highly extensible and can be applied to other data models transparently, including structured relational data models and unstructured key-value pairs. This is made possible by the unclustered architecture on which BIRDB is built, as discussed in [56], which decouples BIRDB's encrypted search index from the underlying storage.

Similar to previous works [53], [64], BIRDB replicates the database D across both the offline and online servers. The database consists of two confidential columns, each containing n keys or values (items). Each item has a size of b bits. Before sending values to the replicas, users encrypt them using their salts to ensure confidentiality during preprocessing. The preprocessed results, also known as hints, can later be recovered by the user using the respective salts. The correctness of this process is guaranteed by the reflexivity property of XOR on boolean operations, as we will explain in §IV.

In line with typical PIR systems [24], [53], the value column in database D is treated as an array, where $D[i]$ represents the i-th value in D. The notation $[x, y]$ is used to denote the set of consecutive integers from x to y, inclusive. For simplicity, when $x = 1$, the notation $[y]$ is used to represent the set $\{1, ..., y\}$.

PIR model (two-phase PIR [23])**.** BIRDB is built on the two-phase PIR which typically comprises four algorithms (`Prep`, `Query`, `Resp`, `Recov`). These algorithms are defined as follows in the setup of BIRDB over a database D of n items:

- `Prep`$(D) \rightarrow h$: this algorithm is executed by BIRDB replicas for each user. It takes the database D as input and outputs a hint h. The hint h consists of three parts: randomly partitioned chunks, filters for chunks, and parities for chunks.
- `Query`$(h, i) \rightarrow q_i$: this algorithm is executed by each user. It takes the hint h and the desired query index i as input and outputs a query q_i.
- `Resp`$(D, q_i) \rightarrow r_i$: this algorithm is executed by the BIRDB master. It takes a query q_i from users as input and outputs a response r_i.
- `Recov`$(h, r_i) \rightarrow d_i$: this algorithm is executed by each user. It takes as input a hint h from a replica and a response r_i from the master, and outputs the desired data object d_i.

These four algorithms work together to enable the metadata-hiding retrieval of specific data objects in BIRDB, ensuring that the access patterns and queried data remain concealed from the untrusted servers.

B. Threat model

In line with recent cloud-based EDBs [21], [24], [53], [64], we assume semi-honest database servers, in the sense that these servers do not collude but are interested in learning the plaintext data and the accessed items. Adversaries, denoted as $\mathcal{A}$, can include database administrators and privileged insiders who can perform passive attacks such as cipher analysis [16] and dictionary attack [18] to extract plaintext information. We rely on the security of one-way functions (OWFs) like SHA-256 [37], assuming that $\mathcal{A}$ is computationally incapable of reversing OWFs or compromising cryptographic encryption keys. These assumptions form the foundation for ensuring the confidentiality of both the data and metadata in BIRDB's encrypted data during encrypted search.

We assume that users are always honest and do not engage in collusion with any servers. While access control mechanisms [58] and blacklists [78] can be used to prevent corrupted clients, we do not consider them within the scope of this work. Additionally, similar to prior research [24], we assume that the underlying storage engine is maliciously secure. This ensures that servers can retrieve the correct data from the underlying storage by detecting rollback and fork attacks, which have been extensively studied in previous storage works [19], [75], [46].

Out-of-scope attacks. We make the assumption that servers do not engage in active attacks such as modifying, dropping, or replaying queries, and that they adhere to BIRDB's protocol. Mitigations exist to authenticate queries and ensure query integrity on plaintext data under active attacks, but addressing integrity under active attacks on encrypted data remains an open problem that we leave for future work [83], [89]. Furthermore, we assume a fixed size for items, which eliminates size leakage. It is challenging to prevent size leakage efficiently without extensive padding to the largest file size (which is costly) or relying on trusted hardware (which requires trust in a third-party vendor). However, mitigations such as partial padding and delayed accessing, where different file chunks are downloaded at different times, can be employed [59], [77].

C. System goals and approaches

At a high level, our goal is to achieve searchability, compactness, and metadata protection for encrypted search, while maintaining the confidentiality of both data and metadata. To accomplish this, our key weapon is a new boolean information retrieval (BIR) technique that allows us to achieve all of these objectives simultaneously. Below, we provide a brief overview of the approaches to achieve these desirable features in EDBs.

Goal 1: Logarithmic encrypted search. Enabling search functionality on encrypted data is crucial for efficient retrieval of specific items without the requirement of decrypting the entire dataset. To address this challenge, our key idea is to replace conventional numerical comparison operations at index branches with secure boolean membership tests. By

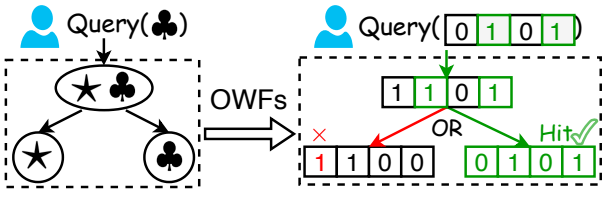


Fig. 4: BIRDB employs boolean information for logarithmic encrypted search by testing membership in index branches and finally comparing every bit at leaf nodes.

incorporating secure boolean information into an orchestrated searchable data structure, we can enable encrypted searches with a logarithmic time complexity.

Figure 4 exemplifies our encrypted search method named BINDEX. This approach involves converting database items into boolean information using cryptographic one-way functions (OWFs), such as SHA-256. Drawing inspiration from the Bloom Filter [13], BINDEX employs a membership-preserving data structure to maintain the membership of the OWF-transformed items, which are represented as binary strings (e.g., '⋆' is represented as '1100'). These binary strings serve as leaf nodes positioned at the lowest index layer, while the inner nodes at higher levels are the *unions* of their child nodes. Each inner node performs an *OR* gate computation on the boolean entries of its child nodes to generate its own boolean information. To search for a specific item, BINDEX compares the '1' bits of the item at each index branch, starting from the root, to ensure a match for every '1' bit. At the leaf node, BINDEX compares each entry, including '0' bits, to ensure an exact match with the queried item.

Goal 2: Compact index with constant overhead. Although achieving logarithmic search complexity (**Goal 1**) is important, it is not sufficient to ensure high performance. This is because as the inner node size of BINDEX grows exponentially when reaching the index root, it becomes challenging to store a large number of index nodes in internal memory. As a result, more nodes are shifted to external storage, which incurs costly I/O operations and hampers performance.

Figure 5 illustrates our approach to compact BINDEX and achieve constant query time and space occupancy. The core idea is that BINDEX's membership information only needs to store the positions of the '1' bits, while the numerous '0' bits are left irrelevant. Taking inspiration from the recent new family of linear approximation model [44], [45], [49], we propose to use an *offset* array to represent the boolean information (specifically, the positions of '1' bits). We then compact these offsets using piece-wise linear functions that map the position in the array to the corresponding offset [31]. For instance, we can utilize two linear functions, defined by only two slopes and intercepts, to represent the offset array belonging to a BINDEX node in Figure 5. Consequently, the implementation of this compacted structure requires constant time and space, regardless of the total number of stored items.

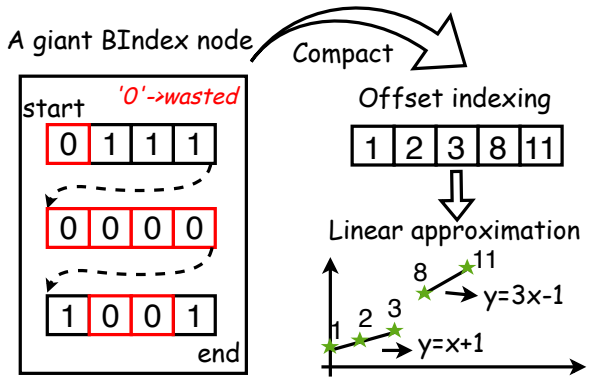


Fig. 5: Based on the boolean information, BIRDB compacts giant index nodes to optimize I/O efficiency and achieve high performance.

Goal 3: End-to-end metadata protection. To effectively conceal metadata and eliminate the leakage of search access patterns, we implement BIRDB using the CK protocol introduced by Corrigan-Gibbs and Kogan [23]. The CK protocol is a cutting-edge two-phase PIR protocol that offloads query-independent computations to a preprocessing phase, resulting in sublinear computation for online queries.

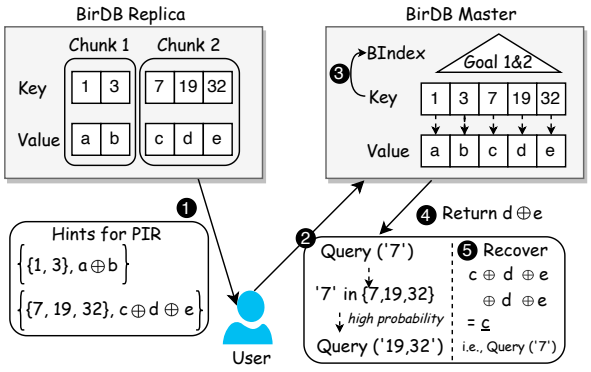


Fig. 6: The third goal of metadata-hiding in BIRDB is achieved through the utilization of two-phase PIR. BINDEX, which accomplishes **Goals 1 and 2**, accelerates the PIR process, while PIR ensures the metadata confidentiality of BINDEX.

Figure 6 demonstrates the integration with two-phase PIR discussed in §III-B. Specifically, the process consists of a preprocessing phase and a query phase. During the preprocessing phase, the BIRDB replica divides the database into random chunks. For each chunk of keys, it computes the parity of the corresponding values and builds filters for each chunk (❶). In the query phase, the user searches for their desired query item by testing its membership in each chunk's filter. Once the target chunk is found, the user deletes the intended query item (❷). The BIRDB master then searches the database using BINDEX based on the query items and computes the parity of the searched values (❸ and ❹). Finally, the user recovers the result by XORing the returned parity with the parity of the original chunk (❺). The correctness of this process is ensured by the reflexivity of XOR on boolean operations; BIRDB accelerates the PIR process using BINDEX (§IV-C).

Overall, by utilizing one core concept of boolean information retrieval, we construct a search index with logarithmic complexity using OWFs. Furthermore, we compact the index by employing linear approximation techniques. Finally, we ensure the concealment of search metadata by leveraging the reflexivity of XOR on boolean operations. Hence, we address all three essential properties: searchability, compactness, and metadata-hiding, in a unified manner.

IV. PROTOCOL DESCRIPTION

A. Desiging an encrypted search index

We first establish the foundation for constructing BIRDB's encrypted search index and subsequently delve into the details of index construction and search methods.

Preliminaries. The underlying searchable data structure in BIRDB is based on the Bloom filter [13], a compact data structure designed for efficient membership testing. The Bloom filter operates by utilizing a single set to store the hash values of all items. Specifically, a Bloom filter B is represented by

an m-bit array and employs k independent and uniformly distributed OWFs, i.e., cryptographic hash functions $\{H_k\}$. Each hash function maps an item to an index in the Bloom filter array. To insert an item x into the filter, we compute its hash using $\{H_k\}$ and set all the bits at the corresponding indices in B to 1. For testing whether an item y is a member of the set, we again hash y using $\{H_k\}$. If any of the bits at the corresponding indices in B are 0, then y is definitely not in the set (i.e., zero false negatives). Conversely, if all the bits at the corresponding indices in B are 1, y is considered to have a high probability of being in the set, albeit with a small configurable false positive rate [13].

Challenges and opportunities. As mentioned above, traditional Bloom filters offer valuable information regarding the *membership* of data for tests. In addition to this, we have observed two key characteristics of Bloom filters that make them promising for encrypted search applications. Firstly, a Bloom filter is *monolithic* in nature, meaning that all items are assigned to a single boolean set. This characteristic allows for efficient item membership tests related to data existence. Secondly, a Bloom filter ensures data *confidentiality* by utilizing one-way functions (OWFs) to transform original plaintexts into hash values of items. The use of OWFs makes the Bloom filter computationally irreversible, effectively hiding the original plaintexts. This security property relies on the assumption of the one-wayness of cryptographic hash functions, which has been extensively studied in prior research [13].

However, despite the valuable membership property and data confidentiality offered by a Bloom filter, the monolithic nature limits its functionality to supporting only membership tests by indicating whether an item (hash) is present in a set. Consequently, Bloom filters alone are not capable of functioning as an index for efficiently searching encrypted data. We observe two significant benefits of using Bloom filters for encrypted search:

• **Security for indexing**. In encrypted search, users do not require the encrypted index keys to be decryptable, as they are used solely for searching corresponding values. Therefore, the utilization of one-way functions (OWFs) to efficiently transform plaintexts into computationally irreversible ciphertexts is sufficient. OWFs provide the necessary security for the index keys without the need for decryption.

• **Compactness**. The boolean nature of Bloom filters allows for the application of cutting-edge compression techniques specifically designed for boolean data. Leveraging these techniques can lead to significant compression gains in the encrypted search index. This is particularly advantageous as traditional encrypted search indexes, such as Order-Preserving Encryption (OPE)-based indexes, are not easily compressible [64], [24]. By reducing the index size through compression, the encrypted search system can benefit from improved performance by avoiding the drawbacks associated with large index sizes and potential performance penalties. We introduce the details of BIRDB compression in §IV-B.

Approach. To construct the encrypted search index, we modify the traditional monolithic Bloom filter by decomposing it into a collection of *minifilters*. Each minifilter is responsible for storing the hash (in bits) of a single item. In other words, a mini-filter represents a Bloom filter dedicated to an individual data item, where only one item (hash) is assigned to the set.

Algorithm 1: Minifilter-based encrypted search

```
1  Function PointSearch(root, key) do
2      if isLeaf(root) then
           ▽ Every corresponding bit should match
3          if key.rft = root.rft        ▷ Minifilter comparison
4          then
5              return root
       ▽ In inner nodes, every '1' bit should match
6      if isInner(root) then
7          if left_minifilter.contain(key) then
8              return PointSearch(root.left, key)
9          if right_minifilter.contain(key) then
10             return PointSearch(root.right, key)
11         return NULL                  ▷ Key is non-existent
12 Function RangeSearch(root, [key1, key2]) do
13     left ← PointSearch(root, key1)
14     right ← PointSearch(root, key2)
15     while left.next != right do
16         value[] ← left.next      ▷ Retrieve sorted leaf nodes
17     return value
```

The minifilters are recursively organized to form a searchable index known as BINDEX. These minifilters have a fixed array size m, and are arranged in a Merkle-tree structure. Each minifilter is stored at a leaf node of the tree, representing an index key. The index keys, such as `"1100"` for record '⋆' in Figure 4, are sorted based on their plaintext key order. The values of the inner nodes, which include all nodes except the leaf nodes, are determined through an element-wise OR gate computation on the minifilters of their child nodes. Let's consider an inner node with l child nodes, denoted as Cn. For each bit (array entry) in the minifilter of the inner node, its value is computed as follows: $Sub_{in}[i] = Cn_1[i]$ *or ... or* $Cn_l[i]$, for all i in the range $[0, m]$. Note that, in the construction of BINDEX, only the leaf nodes in the minifilter tree store actual values, such as record addresses. The inner nodes serve the primary purpose of indexing, facilitating tree traversal for efficient search operations.

Design rationale. The searchability of BINDEX is achieved through the membership-preserving property of the minifilters. Specifically, when any entry in a child node's minifilter is set to `"1"`, the OR gate computation guarantees that the corresponding entry in its parent node's minifilter will also contain a `"1"` bit. For example, in Figure 4, the first entry of the minifilter for '⋆' is `"1"` while the first entry of the minifilter for '♣' is `"0"`. As a result, the first entry of their parent node's minifilter is computed as $1\ OR\ 0 = 1$.

The BINDEX design ensures that if an item exists in a minifilter node, it must also exist in its parent node. In other words, the membership information is propagated to higher-level index nodes. This novel observation, utilized by BINDEX, allows for efficient top-down logarithmic tree traversal based on the membership information. Furthermore, the irreversible confidentiality provided by OWFs directly carries over to the construction of minifilters. By testing the membership of query items in the minifilter nodes, BINDEX eliminates the need for plaintext comparisons typically found in traditional binary search trees (BSTs).

Using BINDEX for encrypted searches. Algorithm 2 outlines the process of an encrypted point query through BINDEX, which involves generating a minifilter for the query item and

traversing BINDEX to locate the corresponding value. BIRDB performs minifilter comparisons at each inner node to verify the membership of each "1" bit in the query item's minifilter. The traversal continues until reaching the leaf nodes, where the match of every bit confirms the presence of a unique index key.

The point search method can be extended to handle range queries. By transforming the range boundaries into minifilters, BINDEX locates two matched leaf nodes using the same method as point queries. All values between these leaf nodes are then returned, akin to the retrieval process in a B^+-tree.

B. Compacting BINDEX *for efficient encrypted search*

A core index challenge: size and cost. Despite BINDEX's efficient tree-based index structure, which facilitates logarithmic search complexity, achieving a low search complexity alone is insufficient for achieving high-performance encrypted search. The reason is two fold. Firstly, as the total number of items increases, each BINDEX node experiences exponential growth in membership information storage due to the membership propagation design (§IV-A). Consequently, the index size significantly expands, adversely affecting search efficiency. Secondly, even in a static setting, the false positive issue, attributed to the fundamental nature of Bloom filters, further amplifies the size of each minifilter node.

To illustrate, let's consider a database with 2^{20} static items and the user aims for a low false positive rate of 1×10^{-3}, same as the configuration in [82], each minifilter in BINDEX would need to be approximately 1.79MB in size using the formula in Bloom filter theory [13]. Consequently, the entire BINDEX would require a substantial amount of storage space, around 3.57TB, which is unaffordable. The significant increase in space occupancy compared to a plaintext database index results in a notable performance decline due to expensive I/O operations involving secondary storage such as disks.

Facing this challenge, our contention is that relying solely on achieving theoretical logarithmic search complexity is insufficient for achieving high-performance encrypted search, even though it represents a significant improvement over linear search methods [22], [25]. Instead, an encrypted search index should adopt a compact design to ensure that the overall query performance remains manageable and cost-effective.

The compact design. As depicted in Figure 9, during index construction, BINDEX preserves light index nodes and compacts giant index nodes that exceed a user-defined memory budget B using two steps below.

• **Step 1 (From bits to offsets).** Recall that in BINDEX, each index node (i.e., minifilter) is represented as a one-item Bloom filter (§IV-A), which is a bit array consisting of only '0' and '1' bits. However, only the '1' bits contain informative data since membership tests only involve checking if all '1' bits are present in a filter. Thus in **step 1**, we extract the positions of all '1' bits from the filter, creating a new offset array. This extraction process allows us to discard unnecessary bits and avoid wasting primary storage resources.

• **Step 2 (Linear Approximation using PGM [31]).** Although **step 1** reduces the size of giant BINDEX nodes, the size still increases exponentially with the total number of records n.

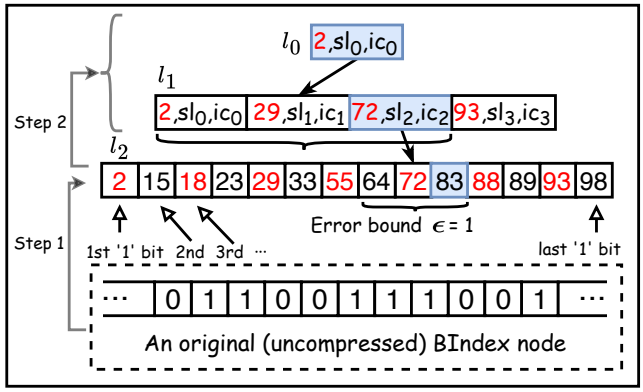


Fig. 7: The construction and a running example that show our compact design for BINDEX. **Step 1** extracts an original BINDEX node into offset representation wherein the ith entry is the position of the ith '1' bit. In **Step 2**, we uses the Piece-wise Linear Approximation model [31] to compact and search the extracted offset array.

Algorithm 2: Internal search of a compressed minifilter

Input:
  A ▷ Sorted offset array of a minifilter node
  ϵ ▷ Error bound of PGM (typically $\epsilon = 1$)
  $levels$ ▷ Level of compacted BINDEX
  k ▷ The query item for search
Function `Query`$(k, A, \epsilon, levels)$ **do**
  $pos \leftarrow f_r(k), where\ r \leftarrow levels[0][0]$
  while $i \leq size(levels)$ **do**
    $lo \leftarrow max\{pos - \epsilon, 0\}$
    $hi \leftarrow min\{pos + \epsilon, size(levels)\}$
    $s \leftarrow []$
    while $s' \in levels[i][lo, hi]$ **do**
      if $s'.key \leq k$ **then**
        $s \leftarrow s'$
    $t \leftarrow right\ most\ segment\ of\ s$
    $pos \leftarrow \lfloor min\{f_s(k), f_t(t.key)\}\rfloor$
  $lo \leftarrow max\{pos - \epsilon, 0\}$
  $hi \leftarrow min\{pos + \epsilon, size(A)\}$
  $return\ binary\ search\ for\ k\ in\ A[lo, hi]$

Fortunately, we find that the cutting-edge compression technique known as Piece-wise Linear Approximation, particularly the PGM [31], is highly suitable for further compressing the extracted offset array in our specific setting.

Specifically, the PGM is a data structure that addresses the linear approximation problem in a multiset of sorted keys within an array A, drawn from a universe U. PGM is parametric with an integer ϵ and employs piece-wise linear approximation models to learn a mapping that returns a position for a given key $k \in U$, which is within ϵ distance from the correct position in A. The fundamental concept behind PGM is that a single linear model, or *segment*, may not be sufficient to ϵ-approximate the positions of all keys in U. Therefore, PGM employs a sequence of segments, each requiring constant space (two floats and one key) and constant query time to provide the ϵ-approximate position of k in A. To construct each segment, PGM reduces the piece-wise linear ϵ-approximation problem to constructing the convex hull of a set of points using a streaming algorithm. This algorithm operates in linear time and space, generating the minimum number of ϵ-approximate segments. We refer to [35], [51], [65] for more on the convex algorithm.

Once the prepared segments, such as the two linear functions shown in Figure 5, are available, the next step is to

identify the ϵ-approximate segment responsible for estimating the approximate position of a query item. To achieve this, PGM begins with the sequence constructed over the entire input array A. It then extracts the first key (indicated by red keys in Figure 9) from A that falls within the coverage of each segment. Subsequently, another linear model is constructed using this reduced set of keys. This recursive process continues until convergence to a single segment. Following the recursive construction, the ϵ-approximate positions returned by the segments can be converted into exact positions through a binary search within a range of $\pm\epsilon$ keys in A. This binary search operation has a time complexity logarithmic to ϵ, regardless of the total number of records in BINDEX's offset array.

Running example.

Complexity analysis: BINDEX's time-space trade-off.

$\mathcal{O}((\log^2 N))$+$\mathcal{O}(m)$ $m \ll n$

C. *Seamless integrating* BINDEX *with PIR*

Overall,

V. EVALUATION

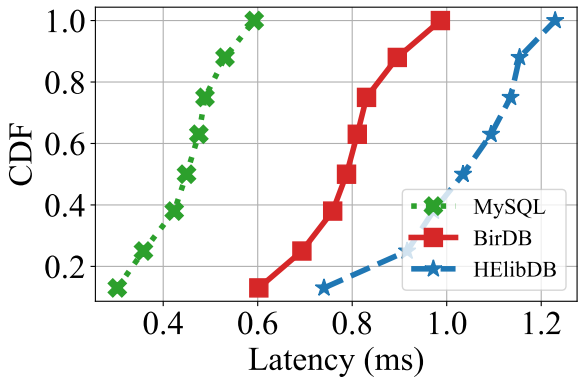


Fig. 8: xxx.

Distributed query latency				
Systems (in SQL instance)	**Original BIRDB**		**HElibDB**	
Query type	PQ	RQ	PQ	RQ
P0 (*client-replica*): Hint preprocess	0.11	0.13	n/a	n/a
P1 (*client*): Parameter transform				
P2 (*master*): Item search				
P3 (*client*): Result Recover				
End-to-end ($\sum P_i$)	0.68	0.73	2.51	3.63

TABLE II: Breakdown 1.

Distributed query latency				
Systems (in SQL instance)	**Original BIRDB**		**Compacted BIRDB**	
Query type	PQ	RQ	PQ	RQ
P0 (*client-replica*): Hint preprocess	0.11	0.13	0.11	0.13
P1 (*client*): Parameter transform				
P2 (*master*): Item search				
P3 (*client*): Result Recover				
End-to-end ($\sum P_i$)				

TABLE III: Breakdown 2.

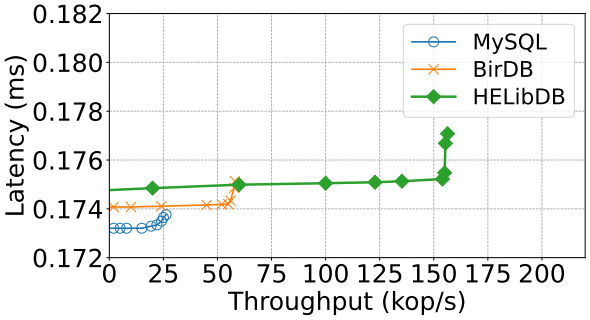


Fig. 9: xxx.

VI. LESSONS LEARNED

VII. CONCLUSION

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